

Exp. No. 5

Design a lexical Analyzer to find the number of whitespaces and newline characters using C.

main.c	Run ▶	Output
<pre>1 #include <stdio.h> 2 int main() { 3 char str[100]; 4 int words = 0, lines = 0, characters = 0; 5 printf("Enter text (up to 100 characters, use ~ to end):\n"); 6 scanf("%[^\n]", str); 7 for (int i = 0; str[i] != '\0'; i++) { 8 if (str[i] == ' ' str[i] == '\t') { 9 words++; 10 } else if (str[i] == '\n') { 11 lines++; 12 } else { 13 characters++; 14 } 15 } 16 if (characters > 0) { 17 words++; 18 lines++; 19 } 20 printf("Total number of words: %d\n", words); 21 printf("Total number of lines: %d\n", lines); 22 printf("Total number of characters: %d\n", characters); 23 return 0; 24 }</pre>		Enter text (up to 100 characters, use ~ to end): Hello World This is C programming~ Total number of words: 5 Total number of lines: 2 Total number of characters: 28